Curriculum Vitae

Tong Wang, M.D., Ph.D.

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Employment

July 2026 — Present Stanford University, Instructor of Flow Cytometry
Physician Scientist Incubator Track
California P&S License: A198934

July 2023 — June 2026 Stanford University, Medical Resident in Clinical Pathology
Chief Resident in Clinical Pathology (2024 – 2025)

Education

August 2015 — May 2023 Perelman School of Medicine at the University of Pennsylvania
M.D.

August 2017 — July 2021 Perelman School of Medicine at the University of Pennsylvania
Ph.D. in Biochemistry and Molecular Biophysics

August 2011 — May 2015 University of Wisconsin—Madison, College of Letters and Science
B.S., Honors in Chemistry

Research Interests

chemical biology, nucleic acids, epigenetics, protein engineering, CRISPR therapy, deep learning, assay development, lab directorship, diagnostic stewardship

Experience

July 2024 – current Department of Genetics, Stanford School of Medicine
Mentor: William Greenleaf, Professor of Genetics
Residency Research
Project: Mutagenesis of *cis*-regulatory elements in hematopoiesis

September 2017 – July 2021 Biochemistry and Molecular Biophysics, Perelman School of Medicine
Mentor: Rahul M. Kohli, Francis C. Wood Professor of Medicine
Dissertation Student
Project: Discovery and application of DNA carboxymethylation

January 2012 – May 2015 Department of Chemistry, University of Wisconsin—Madison
Mentor: Samuel H. Gellman, Ralph F Hirschmann Professor of Chemistry
Undergraduate Researcher
Project: Recognition of α/β peptides by the adaptive immune system

June 2014 – August 2015 MRC Laboratory of Molecular Biology, University of Cambridge
Mentor: Jason W. Chin, Programme Leader
Visiting Researcher:
Project: Synthesis of glycosylated amino acids for genetic code expansion

Preprinted Manuscripts

1) **Wang, T.;** Jessa, S.; Marinov, G. K.; Klemm, S.; Kundaje, A.; Greenleaf, W. J. Sensitive, direct detection of non-coding off-target base editor unwinding and editing in primary cells.
bioRxiv. doi: https://doi.org/10.1101/2025.09.25.678665. *in revision*.

2) Marinov, G. K.; Doughty, B. R.; Schaepe, J. M.; **Wang, T.;** Smith, M. M.; Chen, M.; Kundaje, A.; Sun, Z; Greenleaf, W. J. The human transcription factor occupancy landscape viewed using high-resolution in situ base-conversion strand-specific single-molecule chromatin accessibility mapping. *bioRxiv.* doi: https://doi.org/10.1101/2025.06.27.662080. *in revision.*

First-Author Peer Reviewed Original Research

1) **Wang, T.*;** Park, B.*; Anderson, G.; Shaller, B.; Budvytiene, I.; Banaei, N. Application of Diagnostic Stewardship to Fungal PCR: Low Yield of Follow-up Testing on Plasma and BAL After a Negative Result. *Clinical Infectious Diseases,* **2024.** 79 (4), 944-952. https://doi.org/10.1093/cid/ciae382.

*co-first authors

2) **Wang, T.;** Fowler, J. M.; Liu, L.; Loo, C. E.; Luo, M.; Schutsky, E. K.; Berríos, K. N.; DeNizio, J. E.; Dvorak, A.; Downey, N.; Montermoso, S.; Pingul, B.; Nasrallah, M.; Gosal, W.; Wu, H.; Kohli, R. M. Direct enzymatic sequencing of 5-methylcytosine at single base resolution. *Nature Chemical Biology* **2023.** 19, 1004–1012.* https://doi.org/10.1038/s41589-023-01318-1.

This work describes Direct Methylation-Sequencing (DM-Seq), the first direct, enzymatic, and non-destructive sequencing method for detecting only 5-methylcytosine at base-resolution. I showed that DM-Seq, unlike bisulfite sequencing, unmasks prognostically important cytosines in glioblastomas by not confounding 5mC with 5-hydroxymethylcytosine (5hmC) which have opposite effects on gene expression. This work represents the technological foundation of two patents and a pre-seed diagnostic company ACE Genomics Inc. It additionally describes unappreciated bias in a Base Genomics technology called TAPS that was recently acquired by Exact Sciences for use in liquid biopsy.

News and Views:
Abakir, A.; Reik, W. Direct methylation sequencing. *Nat Chem Biol* **2023**.
https://doi.org/10.1038/s41589-023-01356-9

Highlighted by:
Video Interview, NHGRI Genome Technology Development Coordinating Center:
https://www.youtube.com/watch?v=-Tv80wSS1FI
Genetic and Engineering Biotechnology News:
https://www.genengnews.com/topics/omics/novel-dm-seq-method-profiles-5mc-at-single-base-resolution-using-nanograms-of-dna/
Epigenie:
https://epigenie.com/dm-seq-a-dm-youll-want-to-slide-into-for-direct-methylation-sequencing-of-cytosines/

3) **Wang, T** and Kohli, R. M. Discovery of an Unnatural DNA Modification Derived from a Natural Secondary Metabolite. *Cell Chemical Biology.* **2021,** *28*, 1-8.

This work describes the discovery of the new DNA base 5-carboxymethylcytosine and the enzyme that creates it for the first time. Unexpectedly, I found that only a single point mutation in an endogenous methyltransferase safeguards the *E. coli* genome from creating 5cxmC from the trace metabolite carboxy-SAM (CxSAM). This novel DNA modification was later found by independent groups in naturally occurring viral genomes.

Co-author Peer Reviewed Original Research

1) Fabyanic, E. B.*; Hu, P*; Qiu, Q.*; Berríos, K. N.; Connolly, D. R.; **Wang, T.**; Flournoy, J.; Zhou, Z.; Kohli, R. M.; Wu, H. Joint profiling of 5hmC and 5mC in single cells resolves epigenetic base ambiguity and reveals regulatory strategies of cell-type specific genes by TET dioxygenases. *Nature Biotechnology* **2023**. https://doi.org/10.1038/s41587-023-01909-2.

2) Loo, C. E.*; Hix, M. A.*; **Wang, T.**; Cisneros, G. A.*, Kohli, R. M.* Revealing drivers for carboxy-*S*-adenosyl-L-methionine use by neomorphic variants of a DNA methyltransferase. *ACS Chemical Biology,* **2023.** https://doi.org/10.1021/acschembio.3c00184.

3) Wu, M.*; Shi, L.*; Dubrot, J.*; Merritt, J.; Vijay, V.; Wei, T.; Kessler, E.; Olander, K.; Adli, R.; Tummala, K. S.; Weeresekara, V.; Zhen, Y.; Qibiao, W.; Luo, M.; Shen, W.; Garcia-Beccaria, M.; Fernandez, M.; Hudson, C.; Ronseaux, S.; Sun, Y.; Saad-Berreta R.; Jenkins, R. W.; **Wang, T.**; Heikenwalder, M.; Nicolay, B.; Deshpande, V.; Kohli, R. M.; Zheng, H.; Manguso, R.*; Bardeesy, N. Mutant-IDH Inhibits TET2-Interferon Signaling to Promote Immunoevasion and Tumor Maintenance in Cholangiocarcinoma. *Cancer Discovery* **2021.** 21-1077.

4) Barka, A.*; Berríos, K. N.*; Bailer, P.; Schutsky, E. K.; **Wang, T.;** Kohli, R. M. The base-editing enzyme APOBEC3A catalyzes cytosine deamination in RNA with low proficiency and high selectivity. *ACS Chemical Biology,* **2021,** *17* (3), 629–636

5) Berríos, K. N.; Evitt, N. H.; DeWeerd, R.; Ren, D.; Luo, M.; Barka, A.; **Wang, T.**; Bartman, C. R.; Lan, Y.; Green, A. M.; Shi, J.*; Kohli, R. M.*. Split-Engineered Base Editors for Controllable Genome Editing. *Nature Chemical Biology.* **2021**, *17,* 1262-1270.

6) Caldwell, B. A.; Liu, M. Y.; Prasasya, R. D; **Wang, T.**; DeNizio, J. E.; Leu, A. N.; Amoh, N. Y. A; Krapp, C.; Lan, Y.; Shields, E. J.; Bonasio, R.; Lengner, C. J.; Kohli, R. M.; Bartolomei, M. S. Functionally distinct roles for TET-oxidized 5-methylcytosine bases in somatic reprogramming to pluripotency. *Molecular Cell.* **2021,** *81,* 1-11

7) Ghanty, U; **Wang, T;** Kohli, R. M. Nucleobase Modifiers Identify TET Enzymes as Bifunctional DNA Dioxygenases Capable of Direct N-Demethylation. *Angew. Chem. Int. Ed.* **2020,** *59,* 1-5.

8) Cheloha, R. W.; Woodham, A. W.; Bousbaine, D.; **Wang, T.;** Liu, S.; Sidney, J.; Sette, A.; Gellman, S. H.; Ploegh. H. L. Recognition of Class II MHC Peptide Ligands That Contain β-Amino Acids. *J. Immunol.* **2019.** *203* (6). 1619-1628.

9) Cheloha, R. W.; Sullivan, J. A.; **Wang, T.;** Sand, J. M.; Sidney, J.; Sette, A.; Cook, M. E.; Suresh, M.; Gellman, S. H. Consequences of Periodic α-to-β3 Residue Replacement for Immunological Recognition of Peptide Epitopes*. ACS Chemical Biology* **2015,** *10* (3), 844-854.

Peer Reviewed Scientific Publications (other)

1) Jamal, J. A.; Nicholas, J. A.; **Wang, T.**; Abdelmonem, M.; Pandey, S.; Virk, M. S. Serologic investigation and management of an antibody screen negative para-Bombay phenotype during pregnancy. *Transfusion Medicine* **2025.** http://doi.org/10.1111/tme.70000. (case report)

2) **Wang, T.;** Pizarro-Falcon, S.; Quiros, A.; Bowen, R. A. R. Particulate Matter in Water: An Overlooked Source of Preanalytical Error Producing Erroneous Chemistry Test Results. *Clin Chem Lab Med.* **2024.** https://doi.org/10.1515/cclm-2024-1151. (case report)

3) **Wang, T.** and Silva, O. HHV8+ Diffuse large B-cell lymphoma with EBV coinfection occurring post-transplant. *Blood.* **2024.** 144 (6): 677. https://doi.org/10.1182/blood.2024025072. (case report)

4) **Wang, T.**; Loo, C. E.; Kohli, R. M. Enzymatic approaches for profiling cytosine methylation and hydroxymethylation. *Molecular Metabolism.* **2022***, 57,* 101314. (review)

5) **<u>Wang, T.</u>**; Luo, M; Berríos, K. N.; Schutsky, E. K.; Wu, H.; Kohli, R. M. Bisulfite-Free Sequencing of 5-Hydroxymethylcytosine with APOBEC-Coupled Epigenetic Sequencing (ACE-Seq). *Methods in Molecular Biology* **2021,** 349-367. (book chapter)

Additional Submitted Book Chapter

1) Zhu, J.*; Loo, C. E.;* **<u>Wang T;</u>** Kohli, R. M. Fully enzymatic, Nondestructive Sequencing of 5-Methylcytosine with Direct Methylation Sequencing (DM-Seq). *Methods in Mol. Biol.*

External Oral Presentations

1/21/26—1/24/26	Deaminet, Palm Springs, CA, USA
12/8/25—12/10/25	Nature Cracking the Code: Nucleic Acid Medicines, Boston, MA, USA
8/13/23—8/17/23	ACS National Meeting, San Francisco, CA, USA:
2/9/23—2/12/23	7th Nucleic Acids Fusion Conference, Cancun, Mexico:
1/9/21—1/10/21	APSA Northeast Regional Conference, Virtual
8/17/20—8/20/20	ACS National Meeting, Virtual
11/13/20	Chemistry – Biology Interface Meeting, Philadelphia, PA, USA
7/22/19	Cambridge Epigenetix (CEGX), Cambridge, United Kingdom

Research Funding

<u>Past:</u>

2019—2021	Cambridge Epigenetix (CEGX) Fellowship, $205,620 Title: *Direct methylation sequencing with a DNA carboxymethyltransferase* Location: University of Pennsylvania Role: Trainee

<u>Pending:</u>

	NIH Mentored Clinical Scientist Development Award, K08 Title: *Triaging the impact of off-target CRISPR base edits on human hematopoiesis* Location: Stanford University Role: PI

Awards

2024	Whitlock Prize
2021	Balduin Lucke Memorial Prize
2020	Austrian Basic Science Award
2014	Barry M. Goldwater Scholar
2013	SCORE Program Research Fellow
2013	Hilldale Undergraduate Research Fellow
2011—2015	Wisconsin Academic Excellence Scholarship
2011—2015	Agustin A. Ramirez Jr. Family Foundation Scholarship
2011	Wisconsin Academic Decathlon, First Place Honors Student

Patents

<u>Wang, T.;</u> Schutsky, E, K.; Kohli, R. M. Compositions and Methods for DNA Cytosine Carboxymethylation. 2021. US Patent Application 63027254.

Loo, C. E.; **<u>Wang, T.;</u>** Kohli, R. M. Modified Adapters for Enzymatic DNA Deamination and Methods of use thereof for Epigenetic Sequencing of Free and Immobilized DNA. 2022. US Patent Application 63220650.

Mentoring

2026	Eric Liu (software developer in the Greenleaf Lab)
2024	Gavin Anderson (research specialist in the Banaei Lab)
2023	Saurav Nagpal (undergraduate student in Wu Lab)
2021—2022	Johanna Fowler (research specialist in Kohli Lab)
2020—2021	Laura Liu (high school and undergraduate student in Kohli Lab)
2020—2021	Christian Loo (pre-candidacy graduate student in Kohli Lab)
2019	Bianca Pingul (rotation student in Kohli Lab)
2019	Saira Montermoso (rotation student in Kohli Lab)
2018—2020	Meiqi Luo (research specialist in Kohli Lab)
2010—2011	Lillie Pankratz (high school student in Gellman Lab)

Committees

2024—2025	Department of Pathology Teaching and Training Committee
2024—2025	GME Chief Resident Council
2020—2021	MD-PhD Steering Committee Representative
2018	MD-PhD Retreat Planning Committee
2018	BGS Innovative Ideas Committee Member

Volunteering

2020—2021 The COVID Tracking Project at the Atlantic
Race and Ethnicity Data Collection
Scientific Communication Editorial Team

2015—2016 Health Science Exploration after-school program
Huey Elementary School and Comegys Elementary School

Bylines

Houtman, J., Shultz, L., & **Wang, T.** (2021, February 25). Variants 101: FAQ. The Covid Tracking Project at the Atlantic. https://covidtracking.com/analysis-updates/variants-101-faq

Ledur, J., Rivera, J.M., & **Wang, T.** (2020, September 22). Test Positivity: So Valuable, So Easy to Misinterpret. The Covid Tracking Project at the Atlantic. https://covidtracking.com/analysis-updates/test-positivity

Wang, T., Rivera, J.M., & Ledur, J. (2020, June 29). The Other COVID-19 Metric We Should Be Using. The Covid Tracking Project at the Atlantic. https://covidtracking.com/analysis-updates/the-other-covid-19-metric.

Peer Reviewer (ad hoc)

RSC Chemical Biology, Annals of Family Medicine
Through PhD advisor: *Nature Chemistry, Journal of American Chemical Society, Biochemistry*